

EXP2-4 Experimental Plan P1

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Introduction:

A set of seismic bearings in **EXP2-4** has been determined to be out of specification, prompting a detailed analysis of the processing involved. This report will be based on consultations with the engineering department to identify and assess all potential factors contributing to the bearings falling outside specifications. Finally, design an experimental plan to accurately resolve the specific factors that led to the failure. The plan includes an estimate of the time required for the experiment, the number of samples, the test equipment and the relative resources.

Factors:

Having studied the case, it was determined that three primary factors could have contributed to the building's subsidence and ensuing cracks. Firstly, inadequate molding time and temperature settings, which were thought to be not enough for the large bearings. Furthermore, the change of the accelerator supplier had posed a risk of issues arising from this change. Eventually, the sulphur cross-linker made by the internship student hadn't been examined, probable to induce the trouble.

In the analytical phase, In the analytical phase, the measurement of shear strength and heat resistance of the bearing samples was deemed instrumental in assessing their quality and performance characteristics. Therefore, the experiments following were designed to find out the factors.

Experimental plan:

Among the factors mentioned above, both machine failure and excessive water have the least importance and possibility. Thus, these two factors will be eliminated and ignored to minimize the time cost. To sum up, the time and temperature during the moulding process, the type of the accelerator and the cross-linker amount are regarded as the potential factors that lead to the mechanical property drop of the bearings.

To distinguish different processing steps easily, the standard manufacturing process has been denoted as A, B, and C, where A means the rubber moulding process, B means the accelerator addition process, C means the cross-linker addition process, if there exist some potential wrong operations in these steps, the processing condition will be labeled with a prime symbol (like A', B', and C'). The following experiments will manufacture four types of bearing samples by four different combinations of the three processing conditions(A/A',B/B',C/C').

Experiment 1: A+B+C (All wrong operations that may cause the problem are corrected to get the standard sample.)

Experiment 2: A'+B+C (All wrong operations that may cause the problem are corrected except for A to find whether the problem in A has influence on the final product.

Experiment 3: A+B'+C (All wrong operations that may cause the problem are corrected except for B to find whether the problem in process B has influence on the final product.

Experiment 4: A'+B'+C (Only the operation during the process C is corrected to find whether the problem in process C has influence on the final product.)

Besides these four samples made from experiment 1-4, the original sample made before the accident is also available, recorded as A'+B'+C'.

Then, both the mechanical and thermal testing will be undertaken for the original product and

samples made from the experiments (1-4). In this case, shear strength will be quantified through perforation shear tests, while the glass transition temperature (T_g) will be determined using Differential Scanning Calorimetry (DSC) tests. To reduce potential sources of error, each experiment and measurement will be iterated 5 times to enhance precision and reliability.

After measurement, by comparing the shear strength and T_g of the standard sample with that of the products from experiment 2 and 3, respectively, it can be known that whether the problem during process A (or B) will affect the properties of the earthquake bearings. For example, if the result (values of the shear strength and T_g) from the standard sample is different from that of the sample ($A'+B+C$), it indicates that the wrong operations during moulding process contributed to the property decrease of the bearings. In addition, if the result of the product from the experiment 4 ($A'+B'+C$) are different from that of the original sample made before the accident ($A'+B'+C'$), it indicates that the process C done by Sophie went wrong and also contributed to cause the property drop of the product. To exclude possibility, t-test can be used during the comparison, assuming that the measurement result follows the normal distribution.

It's worth mentioning that Step C is special because Mohammed said that Sophie didn't know what she was doing, so the sample ($A'+B'+C$) should be compared with the initial product rather than the standard sample made from Experiment 1 to find out the result. Figure 1 shows a schematic illustration of the total experiment design.

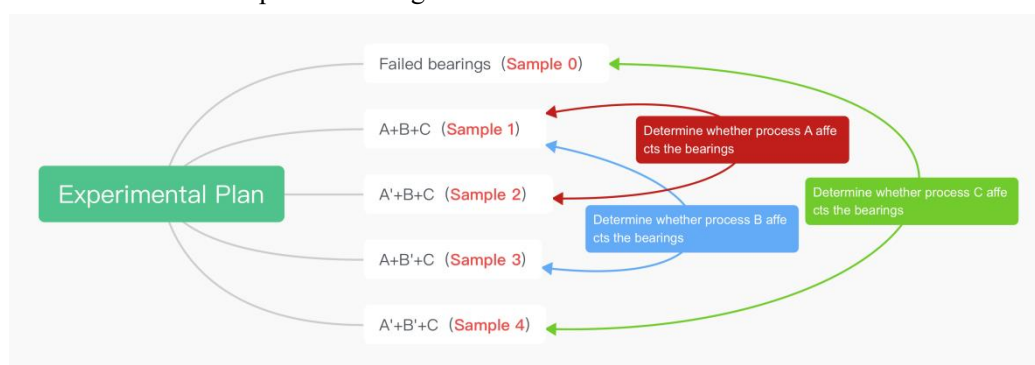


Figure 1. Schematic illustration of experimental plan diagram

Resources estimating:

Equipment: Universal testing machine, DSC, liquid nitrogen

Manpower: All operators need to be trained. To ensure high efficiency, this experiment requires 2 operators, doing the mechanical test and thermal analysis, respectively.

Number of samples: 5 types, 5 samples for each type, 25 samples in total.

Total material consumed: According to GB/T15598-1995, each sample will consume 7.5 cm^3 of material, the total material consumed is 187.5 cm^3 .

Time estimate: The shear strength test for each sample requires 2.5 minutes. DSC test for each sample require about 20 minutes. All samples need a shear strength test, while one sample of each type will be tested by DSC. Taking the operation time into account, about 3 hours are required to finish all tests.

Possible risks:

Operators may be scalded when producing new samples.

Operators may be hurt when cutting the samples.

Toxic gases released during thermal analysis.

Leakage of liquid nitrogen.